RUSSIA

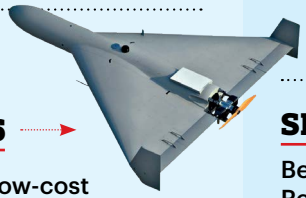
GERAN-5

The most advanced in the Russian Geran series of drones, modelled on Shahed

- **COMBAT RADIUS:** 1,000 km; endurance of 2 hrs
- **WARHEAD:** 90 kg
- **TACTICAL USE:** Heavier payload, improved resistance to jamming. Used for strikes against Ukrainian cities and energy targets

US

LUCAS/ FLM-136



American low-cost attack drone used in waves, reverse-engineered from the Shahed

- **COMBAT RADIUS:** 700-800 km; endurance of 6 hrs
- **WARHEAD:** 18 kg
- **TACTICAL USE:** Used for both swarm strikes on Iranian military targets as well as for counter-swarm functions

INDIAN NEW WAVE

PROJECT KAL



Being designed by IG Defence as a high-precision "deep strike" platform to be used in waves

- **COMBAT RADIUS:** 1,000 km; endurance of 3-5 hrs
- **PRIMARY ROLE:** Precision strikes against fixed high-value infrastructure like command centres and fuel depots

SHESHNAG-150

Being designed by NewSpace Research, it is a sophisticated, AI-driven loitering munition optimised for saturating air defences

- **COMBAT RADIUS:** Over 1,000 km; endurance of over 5 hours
- **PRIMARY ROLE:** Equipped with a 25–40 kg high-explosive warhead, has technology to evade electronic jamming to block satellite signals

enable drone waves to collaborate autonomously. Instead of depending on a central control system, each drone uses its sensor data and communicates real-time with nearby peers, enabling it to change speed or direction to avoid threats while maintaining cohesion.

"India must invest in EW-proof navigation systems, including Inertial Navigation Systems (INS) and terrain mapping capabilities. The objective is to make drones that do not depend on GPS to survive or strike, creating a force that can penetrate defended airspace and execute missions regardless of electronic threats," he explains.

As inspiration, India can look at Iranian drones like Shahed-136 and the smaller Shahed-131 that are designed to operate in such hostile environments. Rather than relying heavily on continuous external inputs, they are built to function with limited or no GPS access. Many of these systems use inertial navigation—a self-contained navigation module—allowing them to maintain course without satellite guidance. A key Indian military official says that Iranian drones ran into some of the same difficulties during the 12-day war in June 2025 that afflicted Indian drones. In eight months, Iran changed the GPS and navigation of its drone inventory, dumping reliance on the US GPS satellite network and switching to China's BeiDou satellite navigation system, making them difficult to track and intercept.

A drone industry insider explains the logic behind Iran's drone strategy. "Low-cost drones can compel defenders to use high-cost interceptor missiles, resulting in an unfavourable economic exchange. Even when defensive systems successfully neutralise incoming threats, the cost incurred can far exceed that of the attacking systems. Modern warfare is no longer defined solely by firepower, but also by economic sustainability," he says.

However, the Shaheds have met their match in the battlefields of Ukraine. Iran had supplied thousands of them to Russia (rebranded as Geran-2 for Shahed-136 and Geran-1 for Shahed-131), and they had entered combat from September 2022. Since 2025, Ukraine has found their antidote—equally low-cost interceptor drones operated by personnel tracking them on a monitor or by wearing first person view (FPV) goggles that are shooting down hundreds of Russian drones. Now, such 'Shahed killers' as the Sting, Bullet and Octopus 100 are being eagerly sought by the US and Gulf countries.



"We are building a mass-deployable strike platform designed for swarm operations—100 to 300 drones launched within a short operational window, overwhelming enemy air defences through sheer saturation. The design is modular and built for rapid manufacturing"

BODHISATTWA SANGHAPRIYA
Founder and CEO, IG Defence

India's Drone Charge

India has several low-cost long-range strike drone projects in the pipeline. The most prominent are the Sheshnag-150, being developed by Bengaluru-based startup NewSpace Research and Technologies (NRT), and Project KAL, by Noida-based IG Defence.

Bodhisattwa Sanghapriya, founder and CEO of IG Defence, says that KAL